CrystalSight: LLMs can read a crystal file, but need geometry tools to reason in 3D

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> **Key finding.** Language models read a crystal file well, but text-only reasoning collapses when a question needs real 3D work. On easy questions (one contact or one plane), CIF text alone scores 0.70–0.82. On hard questions (enumerating symmetry copies, following a hydrogen-bond network, counting neighbours), CIF text drops to 0.20–0.43, and rendered images or an interactive ASCII 3D viewer score 0.10–0.22, near random (0.13). Give the same two models six small geometry tools and both reach 0.75–0.78.

Two sets of spatial questions (20 each, five families of four). The *easy set* (v3) needs one contact or one plane: a donor–acceptor symmetry relation, the H-bond motif, nearest neighbours, a ring's orientation, π-stacking. The *hard set* (v4) needs exact enumeration or graph search, with atoms named in a copy-aware form `SITE#k[i,j,k]` (symmetry operation k, lattice shift [i,j,k]). Its five families are:

- **G1 nearest contact:** the acceptor copy nearest a given hydrogen;
- **G2 H-bond steps:** steps from a molecule to its own lattice translate;
- **G3 H-bond ring:** size of the smallest H-bond ring through a given bond, and molecules it spans;
- **G4 shell count:** molecules within a radius;
- **G5 disorder replicas:** neighbour counts in the ordered major and minor replicas.

Structures come from the Crystallography Open Database and MolCrysKit examples, plus a few licensed CSD entries used locally under anonymous ids and not redistributed.

Gold answers from geometry. Gold answers are computed from symmetry-expanded coordinates. A hydrogen bond is D–H···A with D and A in {N, O}, H···A ≤ 2.5 Å and an angle of at least 120°. Every answer is recomputed with all cutoffs shifted by ±5 %, and questions whose answer changed were replaced. Scoring gives partial credit per JSON field.

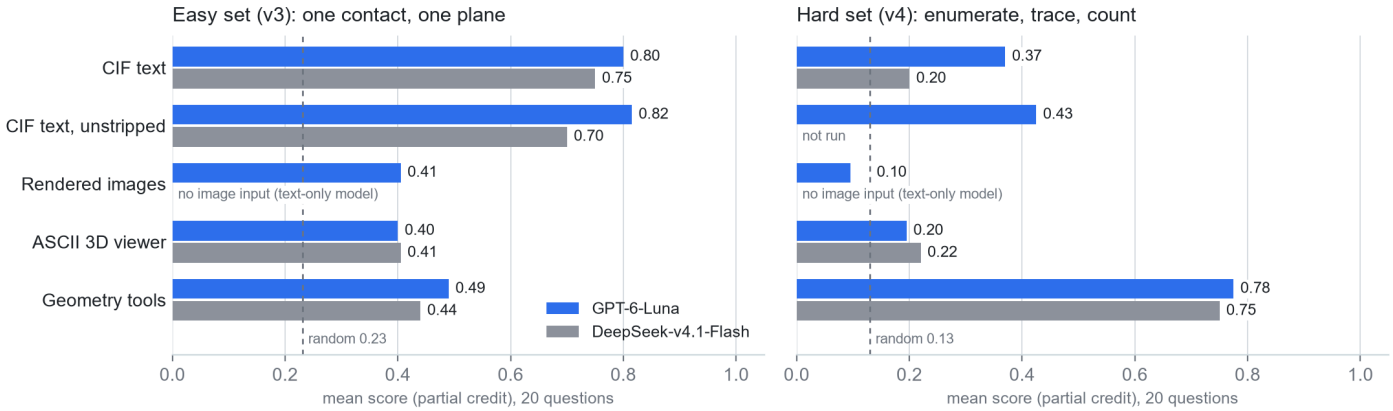
Five channels, one at a time:

- CIF text with publication metadata and answer-revealing hints removed; unstripped CIF as a leakage control; three rendered images;
- an interactive MatterVis ASCII 3D viewer (rotate, zoom, relabel, with an H-bond overlay, distance query and disorder-component switch);
- **geometry tools**: MolCrysKit plus six small tools built for this benchmark (H-bond list, nearest atoms, H-bond path, H-bond ring, neighbour shell, disorder components). The tools return raw lists, which the model must still count and interpret.

Both sets were run with GPT-6-Luna and DeepSeek-v4.1-Flash (text only), at temperature 0 with up to 32k output tokens.

Easy questions: text is enough. Hard 3D questions: only geometry tools hold up

Mean partial-credit score, 20 questions per set. Dashed line = random (0.23 easy, 0.13 hard). DeepSeek CIF (hard): 15/20 empty at 32k-token cap.



Results. On the easy set, CIF text alone scores 0.70–0.82, and images and the viewer about 0.40. On the hard set, GPT-6-Luna's text scores fall to 0.37 (CIF) and 0.43 (unstripped CIF). DeepSeek returned no answer to 15 of 20 CIF questions: it reasoned until the 32k-token limit. Images (0.10) and the viewer (0.20–0.22) stay near chance. With geometry tools, both models reach 0.75–0.78 (strict 0.75). On the easy set, geometry tools score below CIF text (0.49 and 0.44).

The heatmap shows where text survives. G1 nearest contact is solved from text (0.85–1.00). Paths, rings and counts mostly are not, and disorder replicas defeat every channel, tools included (0.25–0.38).

Without tools, only the nearest-contact question (G1) is reliably solved, and only from text

Hard set (v4), mean score per question family (4 questions each). DeepSeek CIF text: old text input, 15/20 replies empty at the token cap.

	G1 nearest contact	G2 H-bond steps	G3 H-bond ring	G4 shell count	G5 disorder replicas
GPT-6-Luna · CIF text	0.85	0.50	0.25	0.25	0.00
GPT-6-Luna · CIF text, unstripped	1.00	0.25	0.63	0.25	0.00
GPT-6-Luna · Rendered images	0.10	0.00	0.00	0.25	0.13
GPT-6-Luna · ASCII 3D viewer	0.60	0.25	0.00	0.00	0.13
GPT-6-Luna · Geometry tools	1.00	0.75	1.00	0.75	0.38
DeepSeek-v4.1-Flash · CIF text	1.00	0.00	0.00	0.00	0.00
DeepSeek-v4.1-Flash · ASCII 3D viewer	0.60	0.25	0.00	0.25	0.00
DeepSeek-v4.1-Flash · Geometry tools	1.00	1.00	1.00	0.50	0.25

Leak found and removed. Many published CIFs include a precomputed hydrogen-bond table with symmetry codes. That table gives away G1 answers and the edges of the G2/G3 network, so for the hard set we remove it from both text channels. We also fixed a text-stripping bug that dropped the `loop_` keyword from CIF tables. GPT-6-Luna's text channels were re-run on the clean input (CIF 0.31 → 0.37, unstripped CIF unchanged at 0.43). DeepSeek's CIF result and all easy-set text results still use the old input.

Limitations. 20 questions per set, 2 models, one run each: one question moves a score by 0.05. The tools were written for these question types, so they show what the right tools make possible, not a general agent. An earlier 247-question set of structure checks (missing H, disorder, coordination), run with two other models, showed the same ordering: text above images and the viewer. Code, tasks, gold and scored results: crystalsight. Geometry: MolCrysKit; ASCII 3D viewer: MatterVis.